



FACULTY OF INFORMATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATICS AND COMPUTER SCIENCE

END OF SEMESTER EXAM

ICS 1201: OBJECT ORIENTED PROGRAMMING I

Date: 27th March 2017

Time: 2 hours

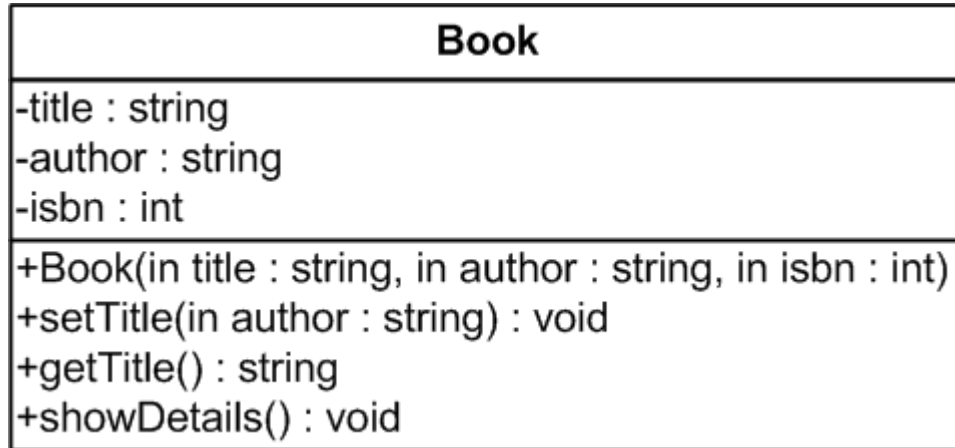
Instructions:

Answer question 1 and any other 2 questions.

Question One [30 Marks]

- a) Discuss the following terms as used in Computer programming **(10 Marks)**
 - i. UML
 - ii. Encapsulation
 - iii. Access modifiers
 - iv. Subclass
 - v. Interface
- b) Explain how data hiding is implemented in Object-oriented design **(6 Marks)**
- c) Using code segment (s), illustrate how object-oriented design implements the hiding of unnecessary details of one class inside another class. **(4 Marks)**

- d) Write a Java class that will implement the Model class shown in the UML diagram below **(10 Marks)**



Question Two [15 Marks]

- a) Using a code segment (s), explain the usage of keyword **super** in Java programming. **(3 Marks)**
- b) Model an inheritance hierarchy of classes Device, Phone and SmartPhone. Class Device has *make* and *weight* as attributes; it also has the set and get methods of all its attributes. Phone has all the properties of class Device, and *manufacturer* as an additional attribute; it also has all the set and get methods for its attributes. Class SmartPhone has all the characteristics of class Phone, and *screen-size* as an additional attribute; it also has all the set and get methods for its attributes. Draw the UML model class diagrams for the three classes described above. **(12 Marks)**

Question Three [15 Marks]

- a) Discuss polymorphism as used in Object-oriented Programming. **(2 Marks)**
- b) Explain of significance of polymorphism in Object-oriented Programming. **(3 Marks)**
- c) John borrowed a Java class from Mary whose UML is shown below. John inherited class Book to create class Novel which has *type* as an additional private attribute. Write a Java class for Novel that will override method *showDetails()* to display the name, author, isbn and type of the book on separate lines. **(10 Marks)**

Book
-title : string -author : string -isbn : int
+Book(in title : string, in author : string, in isbn : int) +setTitle(in author : string) : void +getTitle() : string +showDetails() : void

Question Four [15 Marks]

- a) Explain any difference between the SQL language and the Java language. **(4 Marks)**
- b) Explain any two reasons why JDBC statements are usually placed inside a try-catch structure. **(4 Marks)**
- c) Using a diagram, illustrate the main components of the JDBC architecture. **(7 Marks)**

Question Five [15 Marks]

- a) Explain the main purpose of designing a consistent GUI for any application. **(2 Marks)**
- b) Explain the concept of event-handling as used in Java Programming. **(3 Marks)**
- c) Briefly discuss 5 main GUI design principles you would apply to design a website that aimed at helping Kindergarten children learn basic counting of numbers. **(10 Marks)**